Q Unlike its progress with a Web browser and email client, Mozilla had a brief stint in the operating system space with its Firefox OS. What reason do you ascribe to its failure in the market?

Firstly, although I was an executive sponsor of Firefox OS (codenamed B2G, at first) along with Mike Shaver, we did not get the CEO or other executives fully on board in 2011 when we started. While there was a bigger opportunity in the still-immature Android market, it took us till 2013 to really get to market and, at that point, the window of opportunity was closing due to Google's lead and scale advantages.

Second, even if we had started with full force in 2011, we might have failed, again because of Android's scale and lead-time. A mobile OS requires commodity-scale hardware, deep software integration and, in many parts of the world, a distribution system that has gatekeepers to be paid. It is therefore quite hard for anyone, never mind Mozilla, to do a 'third OS' now. Some say it's impossible, but of course, there are viable Android forks and mods in Asia.

Some day, perhaps, on a device as different from today's smartphones as the first iPhone was from feature phones, there will be a new OS. It certainly needs to be open source for best testing, recruiting and developer ecosystem effects.

Q After JavaScript and Mozilla, you recently expanded your presence in the Web world with Brave Software. Why did you feel the need for a separate Internet security company?

All of the big browser vendors are more or less captured by today's ad-based ecosystem for funding free websites. Apple is the least captured as it walked away from ads as a business (twice, I believe), and so you have seen it add content blockers as an app-install model for extending Safari with ad blocking, starting last fall in iOS 9. But none of the big browsers will block third party trackers and ads, by default. Google would be cutting off its main revenue source if it blocked ads, by default.

Meanwhile, malware in ads (so-called 'malvertising') is coming in through third party ad exchanges. This is just the worst threat, among many—from slow page loads, to drained batteries due to the radio running too much when loading ad-tech scripts and ads or videos. The threats include users being bothered by 'retargeting', overt privacy concerns, and data theft that is legal and does not quite rise to the level of malware. These threats drive users to adopt ad-blockers, which, in the best cases, eliminate bad ads but in general, also deprive websites of revenue from ads.

Brave proposes to fix this broken ecosystem by blocking trackers and ads, by default. It offers users a wallet in the browser, with which to anonymously and effortlessly pay their top sites, and we hope (this is still in the future) even pay users properly for their attention.

We are working now with many small publishers and (soon) with a few large ones to find better ways than ads. Brave cuts out the middle players and proposes several ways for marketers and users to support publishers. Given all the threats and inefficiencies of online advertising and the downward trends, we believe the Web is ready for a new set of standards for anonymous and private ads and micropayments.

Our long term value is not in how we innovate to create just some future standards, rather it is our users' trust in us putting them first, always giving them the same or even larger cut of any revenue based on their attention, and never holding our users' data in the clear off of each user's devices. All this means we never hold cleartext user browsing data on our or anyone else's servers.

Q How is the Brave browser different from Firefox, despite both being open source offerings?

Brave blocks third-party ads and trackers, by default, and works hard for user privacy in other ways (HTTPS everywhere activated, by default, and fingerprinting protection). Other browsers, including Firefox, do not block adverts by default, because they do not want to rock the boat.

Brave also differs by being based on Electron and (same as Chrome's engine) Chromium. This gives us the market-designated best engine with the most widely used extensions, but not any of the Google accounts or other service integrations, nor the front-end features of Chrome, which we do not want. Brave has its own user interface, built via Electron using React.js, HTML, CSS, SVG and images.

For users, particularly mobile users, the speed with which Brave loads Web pages directly translates into benefits. For instance, Brave Android users will see a 2x to 4x speed increase, which correlates with battery and data plan optimisation.

Q Lastly, how do you see the parallel journeys of open source and the Web?

Canadian science-fiction author Cory Doctorow in his recent talk on the darker view of things notes that while open source has won, the Web is in trouble due to closed source and worse threats such as DRM and the laws enforcing it. I believe, open source, which goes back a long way to when IBM released its Fortran, is inevitable in any broad, commercial-in-part ecosystem that has stable kernels across evolutionary time, transforming from UNIX to Linux, TCP/IP, HTML and JavaScript. These kernels and analogues of them in other layers of the ecosystem's 'stack' become the cost rather than profit centres, over time. The Web now has over three giant open source engines backed by multi-million LOC (lines of code) projects, namely Mozilla's Gecko, the Apple-led WebKit and the Google-led Chromium. Microsoft has even recently open-sourced its new ChakraCore JavaScript engine.

However, I see not so much a parallel as a symbiosis. The Web is big enough and commercialised now, so inevitably its evolutionary kernels will only be open-sourced for best peer-review effects such as development, QA and evangelism. Open source, in turn, symbiotically promotes open standards by providing reference implementations an A/B whitebox testing which is better than any closed-source approach that feeds into standards. END